

Smart India Hackathon 2026

AI-Based Compliance Monitoring for Skill Training Centres

*Verifying that trainees are really present and approved equipment is really available,
with or without CCTV, while protecting privacy*

Final Detailed Solution Document

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1. The problem and the scenario

Government skilling schemes pay training centres based on how many trainees attend and what infrastructure the centre provides. Centres are spread across thousands of villages and towns and are physically inspected only now and then. Between inspections, two kinds of leakage happen:

- **Ghost or proxy attendance:** a trainee who never comes to class is marked present, or a friend marks attendance for them.
- **Missing infrastructure:** the approved lathe, workbenches or computers are not there, are broken, or are only brought out on inspection day.

The scenario we are solving

Imagine a vocational centre in a semi-urban area. Its register says **30 trainees attended today**, but only **12 were actually in the room**. The centre has **no CCTV camera**. Some trainees mark attendance from home, and one hands his phone to a friend to mark for him. The sanctioned list says the workshop has 2 lathes, but one has been broken for weeks. Nobody finds out until the next scheduled inspection, months away.

Other centres do have CCTV, but it only records passively. Nobody watches the footage, and it is not connected to attendance records at all.

What we need to build

A system that works with or without cameras, proves a person was physically inside the classroom, cross-checks that against what the centre reports, flags gaps for a human officer, works on poor internet, and respects trainee privacy.

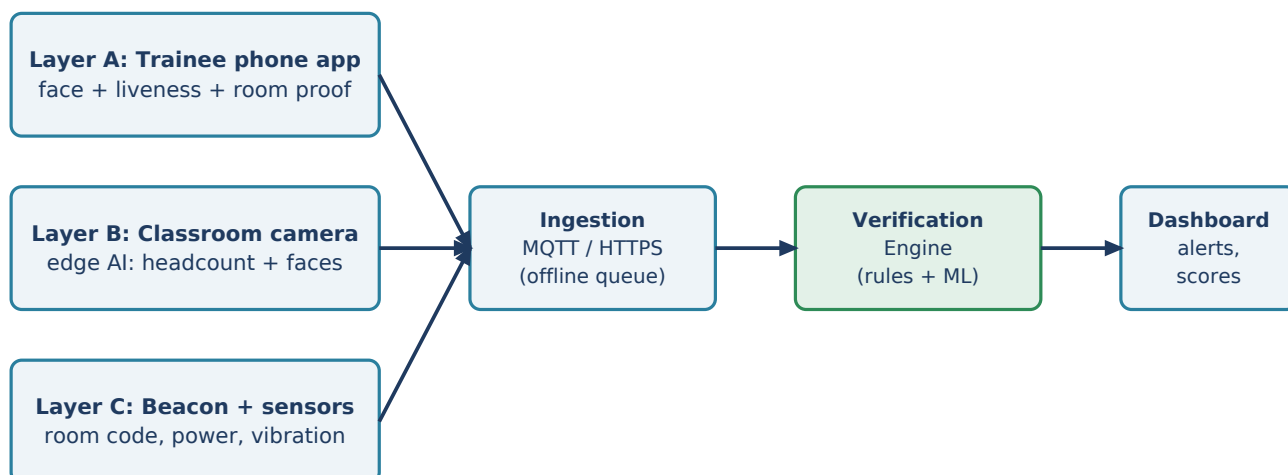
2. Our solution in one page

Core idea: trust, but verify. Every attendance mark must be backed by independent proof, and every claim by the centre is compared with evidence that the centre cannot easily control.

Three kinds of centres, one system

Centre type	How presence is proven	Why this works
No camera	Trainee phone app: live face check, liveness, and proof the phone heard the classroom's secret code	Needs only phones the trainees already have and one cheap beacon in the room
Camera only	Camera at the centre counts people and matches faces of enrolled trainees against their own batch	Uses infrastructure that already exists, so no new cost
Camera and phone (best)	Both are used and cross-checked against each other and against the centre's register	Two independent sources agreeing gives the highest confidence

The three layers



Only numbers and small blurred thumbnails travel upward. Raw video never leaves the centre.

- **Layer A: Phone app.** Proves the right person, alive, in the room.
- **Layer B: Camera and edge AI.** Counts people and recognises enrolled trainees, on a small device inside the centre.
- **Layer C: Beacon and sensors.** Provides the room-only code and shows whether machines are really running.
- **Verification engine.** Compares all the evidence with the centre's claims and produces one of three outcomes: Verified, Needs review, Suspicious.
- **Dashboard.** Shows the monitoring unit only the exceptions, so no one needs to watch raw video.

3. How this answers each requirement of the problem statement

Problem statement asks for	Our answer
Estimate actual attendance from camera feeds and cross-check with the centre's records	Edge AI counts people and matches enrolled faces, then the engine compares them with the register and the phone marks
Detect presence and working condition of approved infrastructure against sanctioned inventory	Object detection against the centre's sanctioned list, plus power and vibration sensors for whether machines actually run
Flag discrepancies on a dashboard, not raw footage	Flags go to the monitoring dashboard with a small blurred thumbnail, evidence and a confidence level
Work within real bandwidth and camera quality in rural centres	All AI runs locally. Only about 1 KB per event is sent. Works offline and syncs later. Low-light and low-resolution handling is built in
Preserve privacy: aggregate presence where individual ID is not needed	Headcount is the default. Faces are matched only for enrolled and consenting trainees, and only embeddings (numbers) are stored, never photos or video
Scenario: centres without cameras	Phone-based attendance with room proof, plus random trainer photo checks for equipment

4. A worked example: one class, four trainees

Centre register: **30 present**. Let's follow four trainees through one attendance prompt.

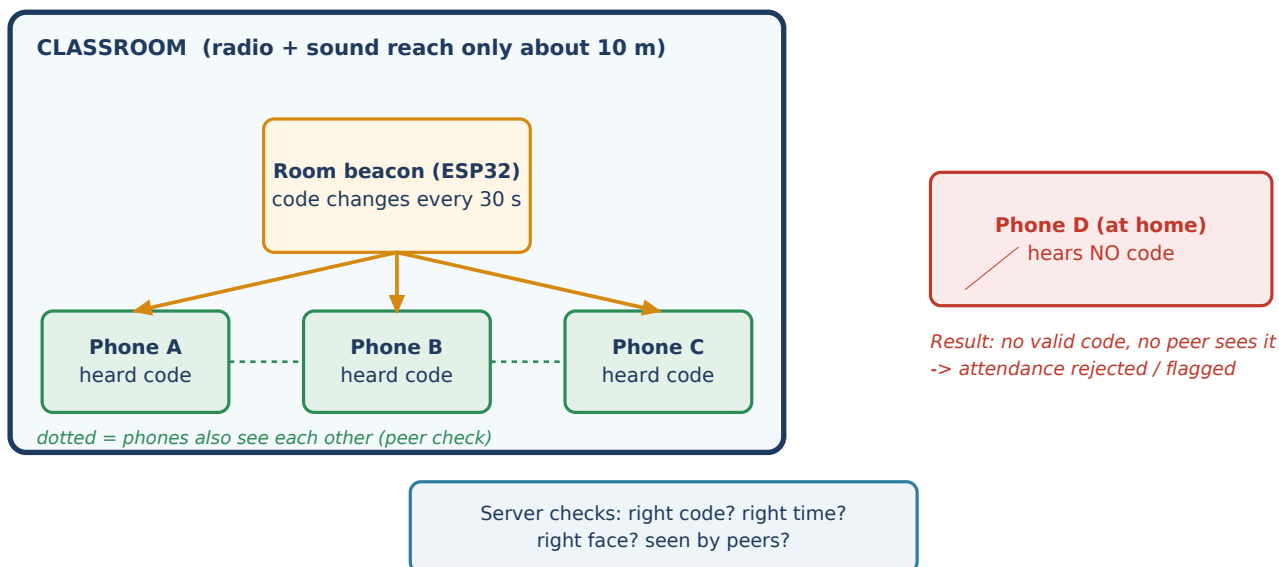
Trainee	What happened	What the system saw	Result
Ravi	In class. Opens the app when the prompt arrives	Heard the room code, live face matched, other phones saw his phone, camera also saw him	Verified
Amit	Absent. Gave his phone to a friend inside the room	Room code heard, but the friend's face did not match Amit's. Liveness challenge passed but identity failed	Suspicious
Sunita	Absent, at home. Tries to mark from there	No room code reachable from home. No peer phone saw her phone	Rejected / Suspicious
Mohan	In class, but phone battery died, so no app mark	No app mark, but the camera saw and recognised him	Missing mark (flagged for correction, not penalised)

At session level: the centre claims 30. The app shows 12 verified and the camera counts 12 to 13 people. The engine raises a **session discrepancy** and sends it to the district officer's dashboard, with the evidence, instead of waiting for the next inspection.

5. How we prove a trainee is physically inside the room

This is the most important part, so here it is in plain language.

A phone cannot be trusted when it says "I am in the classroom", because GPS can be faked. So we place something in the room that only a person inside can hear, and the phone must prove it heard it.



The room beacon

A small device (an ESP32 board, roughly ₹400) is fixed on the classroom wall. It broadcasts a **6-digit code over Bluetooth that changes every 30 seconds**. Only the beacon and the server know the pattern behind the code (this method is called TOTP, the same idea as an authenticator app). Optionally, it also plays the code as a **soft sound token**, which the phone microphone must hear. Sound does not pass through walls as easily as radio, so it helps against someone standing just outside.

The five proofs, and why each one exists

Proof	What it does	Why we use it	Technology
1. Room code	Phone must hear the current code from the beacon	Marking from home fails because the beacon is out of range. A forwarded code expires in 30 s	ESP32 BLE beacon, TOTP, optional sound token
2. Face match	Compares the live face to the trainee's enrolled face	Stops 'phone given to a friend'	MobileFaceNet or ArcFace on the phone (TFLite)
3. Liveness	Random challenge (blink twice, turn left) plus a passive photo/screen detector	Stops a photo, a video or a deepfake held to the camera	ML Kit, MediaPipe, MiniFASNet anti-spoofing
4. Peer proof	Phones in the room detect each other over Bluetooth	A phone that no other phone ever saw is probably not in the room	BLE peer scanning
5. Genuine phone	Detects rooted phones, emulators, fake GPS, tampered apps; signs data with a hardware-protected key	Stops software tricks and forged submissions	Play Integrity API, Android Keystore, signed payloads

Step by step: one attendance check

- The server sends a **push notification at a random time** (2 to 3 times per class, 5-minute window). So nobody can mark once and leave.
- The student opens the app. The app **listens** for the room code and checks the phone is genuine.
- The app asks for a **random live challenge** and matches the face with the stored template **on the phone**.
- The app creates a **signed packet** and sends it. If there is no internet, it saves the packet and sends it later.
- The server checks the code, time, signature, face score and peer sightings, and gives the check-in a **trust level**.

```
{ "trainee_id":"T123", "session_id":"S45", "ping_no":2,
  "room_code":"482913", "face_match":0.91, "liveness":0.97,
  "integrity":"PASS", "peers_seen":["T118","T120"],
  "ts":"2026-10-01T10:14:03Z", "sig":"..." }
```

One-time enrolment

At admission the trainee verifies identity with the scheme ID (Aadhaar eKYC where permitted) plus OTP, gives consent, and takes a selfie. The app converts it into a **face embedding** (a list of numbers) and stores that, not the photo. The account is **bound to one device**. Changing phones needs admin approval.

6. Every cheat we thought of, and how it is handled

Cheat	Why it fails / how we respond
Phone handed to a friend in class	The friend's face does not match the enrolled face. Repeated failures raise an alert
Marking from home	Beacon code cannot be heard from outside Bluetooth range. Peer phones never see this phone
Standing just outside the door	Sound token is weak through walls, peer proof and camera headcount disagree. Several signals must agree
Code forwarded on WhatsApp or screenshot	Code expires in 30 seconds and the submission is signed by that phone for that trainee
Fake GPS app, emulator, rooted phone	Device integrity check fails. GPS is only a minor extra signal anyway
Photo, video or deepfake of the face	Random live challenge plus anti-spoof model
Mark once, then leave	2 to 3 random check-ins per class. Attendance is the fraction passed
Two phones for one trainee	One trainee is bound to one device
Centre edits its own register	The register is compared with independent app, camera and sensor evidence
Trainer colluding with the centre	Camera counts and sensor data are independent of the trainer. Anomalies go to the district and state, not only to the centre
Relay of Bluetooth signal from outside	Hard, because several independent proofs must agree. We state this as a known limit (Section 17)

7. Camera path: for centres that already have CCTV

A small **edge device** sits inside the centre, reads the camera stream, and processes it there. Only results leave the centre.

Camera stream -> take 1 frame every 5-10 s -> improve low light (CLAHE)
 -> detect people (YOLO) -> track each person (ByteTrack) -> headcount
 -> detect faces (SCRFD / RetinaFace) -> keep only sharp faces
 -> face embedding (ArcFace / AdaFace) -> match against THIS batch only (~30 people)
 -> identity + confidence, or 'unknown'

Step	What it does	Why we do it this way
Frame sampling	Looks at one frame every few seconds, not every frame	Saves computing and power. Attendance does not need 30 frames a second
Person detection and tracking	Counts people and follows each one so nobody is counted twice	Gives an identity-free headcount, which is the privacy-friendly default
Face quality filter	Throws away blurred, tiny or side-on faces	Wrong guesses are worse than no guess. Poor faces are marked 'not confirmed'
Average over many frames	Combines the same person's face across several frames	One frame from CCTV is unreliable. Combining many is far more stable
Small gallery (this batch only)	Matches against about 30 enrolled people, not a big database	Fewer candidates means far fewer false matches, and it limits data exposure
Unknown rejection	Visitors and staff get 'unknown'	Prevents forcing someone onto a trainee's name
One person, one identity	Two people cannot both be assigned the same trainee	Removes duplicate matches
Fallback for hidden faces	Body appearance and seat position as a weaker clue	Faces turn away or are hidden. Used with lower confidence only

Honest note: published accuracy figures for face recognition come from clean photos. Classroom CCTV is harder (low light, angles, small faces). We therefore report only our own measured results on

our test footage (Section 15) and use quality filtering to avoid wrong matches.

8. Equipment and infrastructure verification

Each centre has a **sanctioned inventory**, for example 10 workbenches, 2 lathes, 20 chairs, 15 computers. The system checks three things: is it **there**, is it **in place**, is it **working**.

Check	Method	Why
Is it there?	Object detection (YOLO or RT-DETR fine-tuned on workshop images) counts benches, machines, chairs and computers, then compares with the sanctioned list	Direct answer to 'expected 10, found 6'
Has the layout changed?	Compares with a baseline reference photo of the approved layout (change detection)	Catches equipment that has been moved out or swapped
Is a machine really working?	Current sensor or smart plug (PZEM-004T), plus vibration or sound sensor	A lathe 'used for 6 hours' that drew no current is a clear discrepancy. This is far more reliable than looking at a still picture
Is a computer on?	Screen-on detection in the image, optional usage logs	Basic operability check
No camera at the centre	Trainer is prompted at a random time to take a live photo inside the app (no gallery upload) with timestamp, location and device check. The same detector checks it	Keeps equipment checking possible without CCTV, and a missed prompt is itself a flag

The result for each item is shown as **expected / detected / operable**, with a confidence level, so officers can see exactly what is missing.

9. The decision engine: cross-verification

For every session, the engine collects four kinds of evidence and compares them with what the centre claims:

- **M**: the centre's register and the app marks
- **C**: camera evidence (headcount and recognised enrolled trainees)
- **B**: beacon, sound and peer proof
- **S**: sensor evidence (power, vibration)

Situation	Outcome
Marked, with room code, live face match, peer sighting (and camera sighting where there is a camera)	Verified
Evidence is partial or borderline	Needs review
Marked, but no room proof, or face mismatch, or not seen by camera or peers	Suspicious
Seen on camera but not marked	Missing mark (correct the record, no penalty)
Register says 30, evidence supports 12	Session discrepancy
Centre register differs from app records	Record mismatch
Equipment missing, or machine shows no running signal	Infrastructure flag

How we keep the engine accurate and fair

- **Two-signal rule.** One weak signal never raises a flag. At least two independent signals must disagree.

- **Tolerance bands.** A session is flagged only when the gap exceeds a margin (for example more than 3 people or about 15%), because cameras can miss people behind others.
- **Learned scoring.** Instead of hand-picked weights, we train a simple model (logistic regression or gradient boosting) on our labelled test sessions and calibrate the score so that '0.9' truly means about 90%.
- **Pattern detection over time.** An Isolation Forest looks for oddities such as identical timestamps from many phones, or a centre that shows 100% every day.
- **Human in the loop.** The system never penalises anyone automatically. It flags. An officer decides, using the evidence.
- **Confidence always shown.** Every record carries a confidence level, so the system never pretends to be certain.

10. What if many students mark attendance at the same time?

It works fine. A class of 30 is a tiny load, and the design handles bursts.

Concern	How it is handled
Will one shared room code confuse the system?	No. The code is like a clock on the wall, shared by everyone. Each phone signs the code together with its own trainee ID and device key, so each submission is unique and cannot be reused by someone else
30 students tap at once	Each student gets a 5-minute window and submissions are spread out. A queue (Redis/Celery) absorbs any burst, so nothing is lost
Double taps or network retries	One mark per student per prompt. Every request has a unique ID, and the database has a unique rule on (student, session, prompt), so duplicates are ignored
Weak network	The signed mark is saved on the phone and uploaded later . Signed timestamps stop backdating
Many phones scanning Bluetooth	Bluetooth is designed for many devices. Phones scan in short bursts to save battery, and one missed signal does not flag anyone because we judge the whole class session
Scale to the whole country	Even 10,000 centres with 30 trainees is about 300,000 check-ins spread over minutes. This is comfortable for a standard FastAPI and PostgreSQL setup that can add servers

A crowd also does not hide a cheat. Every phone is checked individually, so one student marking for an outside friend is still caught by the face check, the peer check and the camera headcount.

11. The monitoring dashboard

The dashboard is for district, state and ministry officers. It replaces watching footage with a short list of things that need attention.

- **Map view:** every centre coloured by compliance level, so problem areas stand out
- **Centre page:** claimed versus detected attendance over time, trust distribution, and the equipment table (expected / detected / operable)
- **Alert feed:** ranked flags, each with evidence, a blurred thumbnail, and a confidence level
- **Follow-up workflow:** assign to an inspector, record the outcome, close or escalate
- **Reports:** monthly centre and district compliance summaries for the ministry
- **Access control:** separate views for ministry, state, district and centre users, with a log of who viewed what

12. Privacy by design

The problem statement asks us to prefer aggregate detection wherever identity is not needed. Our principle: **headcount is the default, identification is the exception.**

Question	Our answer
What do we detect by default?	How many people are present, not who they are
When do we identify someone?	Only an enrolled trainee who consented, to verify that trainee's own attendance claim
Who is never identified?	Staff, visitors and bystanders. They are counted in aggregate and shown as 'unknown'
What is stored?	Face embeddings (encrypted numbers), counts and scores. No photos and no video are stored on our servers
Where is face matching done?	On the trainee's phone for app check-in, and on the edge device in the centre for cameras
What leaves the centre?	Small data packets, plus blurred thumbnails only when something is flagged
How long is data kept?	Embeddings deleted at course end or on request. Flag thumbnails kept for a limited period (for example 30 to 90 days)
What law and standards guide this?	India's DPDP Act 2023: consent, purpose limitation, data minimisation. Encrypted storage and transport, with access logs

For the pitch: have the exact consent wording and retention periods confirmed by the scheme's legal or data-protection team before deployment. We present these as design choices, not legal advice.

13. Low-bandwidth and offline operation

- **No video upload.** AI runs on the phone or the edge device. Only results are sent.
- **Small messages.** About 1 KB of JSON per event over a lightweight protocol (MQTT), batched.
- **Offline first.** Events are stored on the device and synced when the network returns. They carry signed timestamps so they cannot be backdated.
- **Compressed models.** AI models are quantised (INT8) so they run on cheap hardware and old phones.
- **Small thumbnails.** Flag images are blurred and compressed to roughly 15 to 20 KB.
- **Low-quality cameras.** Frame sampling, brightness correction, and quality filters keep results usable at low resolution.

We will **measure the real data used per centre per day** during the demo and quote that number, rather than only an estimate.

14. Technology stack: what each tool does and why we chose it

Area	Technology	What it does here	Why we chose it
Mobile app	Flutter	Single app for trainees and trainers	One codebase, fast to build, good Bluetooth and camera support
On-phone face and liveness	TFLite, MobileFaceNet / ArcFace, ML Kit, MediaPipe, MiniFASNet	Face match and anti-spoof on the phone	Runs offline, keeps face data on the device
Phone trust	Play Integrity API, Android Keystore	Detects fake or tampered phones, signs data	Hardware-backed and hard to forge
Room proof	ESP32 board, BLE, TOTP	Broadcasts the rotating room code	Very cheap, low power, easy to deploy
Camera AI	YOLO (v8/11), ByteTrack	Counts people and equipment, tracks without double counting	Fast, accurate, runs on small devices
Face recognition (camera)	InsightFace: SCRFD / RetinaFace + ArcFace / AdaFace	Detects and matches faces to the batch	Widely used strong open models
Edge hardware	Mini PC or Jetson-class device, OpenVINO / TensorRT, ONNX	Runs AI inside the centre	Processing stays local, saves bandwidth, protects privacy
Sensors	PZEM-004T current sensor, smart plug, vibration or MEMS mic	Shows whether machines are actually running	Cheap and objective proof of use
Messaging	MQTT (Mosquitto)	Sends small events from centres	Built for weak networks, supports offline
Backend	FastAPI, PostgreSQL (PostGIS), Redis, Celery	APIs, storage, queues, scheduled prompts	Reliable, scalable, easy to maintain
Evidence storage	MinIO / S3	Holds blurred flag thumbnails	Low cost, access controlled
Decision models	scikit-learn, XGBoost, Isolation Forest, MLflow	Learns scoring, finds anomalies, tracks experiments	Simple, explainable, easy to audit
Dashboard	React, Leaflet, Recharts (Streamlit for the prototype)	Map, alerts, charts, workflow	Fast to build clear officer views
Security	TLS, AES-256, JWT with role-based access, audit logs	Protects data in transit and at rest, controls who sees what	Meets expected government security practice
Deployment	Docker Compose	Packages the system	Same setup on any server

Android first. Bluetooth peer scanning and Play Integrity work best on Android, which most trainees use. iPhone support would need extra work and has different background Bluetooth limits. We state this as a pilot scope decision.

15. Accuracy and how we control errors

Because we cannot claim accuracy without evidence, we **measure our own results** on a demonstration dataset and publish them honestly.

Demonstration dataset

8 to 10 simulated class sessions filmed with phones or a webcam, using our team and friends, plus public classroom clips. We **deliberately inject cheats**: absent trainees marked present, a phone handed to a friend, a code shared by screenshot, marking from outside the room, and equipment removed or switched off. Every session has manual ground truth.

What we measure

Component	Measures	What it tells us
Face recognition	True accept rate at a fixed false accept rate, EER	How often the right person passes and the wrong one is rejected
Liveness	APCER and BPCER (ISO 30107-3 style)	How well fake faces are rejected without blocking real users
Headcount	Mean absolute error, percentage error	How far the people count is from the real count
Equipment detection	mAP, per-item precision and recall	Whether benches, machines and chairs are found correctly
Proxy and discrepancy flags	Precision, recall, false-positive rate, false-negative rate, confusion matrix	How many flags were right, and how many cheats were missed
Conditions	Results split by low light, crowding and hidden faces	Where the system is weak, shown openly

Result table to fill from our runs	Our measured value
Headcount error (MAE)	<i>to be measured</i>
Proxy detection: precision / recall	<i>to be measured</i>
False-positive rate (honest trainee wrongly flagged)	<i>to be measured</i>
False-negative rate (cheat missed)	<i>to be measured</i>
Equipment detection precision / recall	<i>to be measured</i>
Data sent per centre per day	<i>to be measured</i>

Built-in error control

- Quality filtering, so a blurred face is never forced into a match
- Multiple independent signals, so a single mistake cannot flag a trainee
- A review band for borderline cases, plus tolerance margins on headcount
- A trainer-assisted fallback with lower confidence for old phones or bad lighting
- A threshold sweep (ROC curve) so we can show the trade-off between false positives and false negatives and choose where to operate

16. Cost and rollout

Item	Indicative cost	Where used
Trainee app	Free to trainees	All centres
Room beacon (ESP32)	About ₹400	Every classroom, especially centres without cameras
Current sensor / smart plug	About ₹500 to 800 per machine	Key machines in workshops
Edge device for camera centres	From a few thousand rupees (low-cost board) to a higher-priced AI device, depending on how many cameras and how much processing	Only centres with CCTV

Prices are indicative and should be confirmed against current market rates. The one-time hardware cost can be framed as part of **empanelment requirements**, paid back through reduced leakage.

Phased rollout

Phase	What happens
Pilot	A few centres in one district. Measure accuracy, battery, bandwidth and trainee acceptance
Tune	Adjust thresholds with real data, fix usability issues, finalise privacy and consent process
Scale	State-wide rollout, with centres grouped by what equipment they have
National	Integrate with the scheme's existing reporting systems and ministry dashboards

Build plan for the hackathon

Days	Work
1-2	Problem framing, architecture, dataset collection, backend skeleton
3-5	Mobile app: room code check-in, liveness and face match. ESP32 beacon
6-8	Camera AI: headcount, face match, equipment detection. Edge script and messaging
9-10	Decision engine, scoring, anomaly detection
11-12	Dashboard, alerts, sensor demo
13-14	Accuracy testing, privacy note, low-bandwidth test, demo video

17. Honest limitations

- **Relay attacks.** A skilled cheat could try to relay the Bluetooth signal. That is why we need several independent proofs, surprise timing, and camera or peer confirmation. We make this costly and detectable, not impossible.
- **Poor lighting or old phones.** Face match can fail. The fallback is trainer-assisted with lower confidence and goes to review, not accusation.
- **Camera limits.** Accuracy drops with occlusion, crowding and low resolution. Reported separately in our results.
- **Sound token.** Needs microphone permission and must be tested in noisy workshops. It is an optional extra, not the only proof.
- **Hardware cost.** Needs a small one-time investment per centre.
- **Not a replacement for inspectors.** It tells inspectors where to look.

Likely judge questions

Question	Answer
What if a trainee has no smartphone or a very old one?	Trainer-assisted check with lower confidence. The record goes to review, not to suspicious
Can Bluetooth be relayed?	Yes in theory. We require multiple independent proofs, random timing and camera or peer confirmation, so cheating becomes costly and detectable
Who is responsible if the AI is wrong?	No automatic penalty. A human officer checks the evidence for every flag
Is face recognition a privacy risk?	Default is headcount only. Matching is on-device or at the edge, only embeddings are stored, with consent and deletion rules
What if the centre has no internet?	Everything works offline and syncs later with signed timestamps
What if the centre has no camera?	Phone app with room proof, and random trainer photo checks for equipment
What does it cost?	About ₹400 per centre for the beacon in the basic setup, plus sensors and an edge device only where needed

"We don't promise cheating is impossible. We make it detectable, costly, and always backed by evidence, so officers inspect the right centres at the right time."